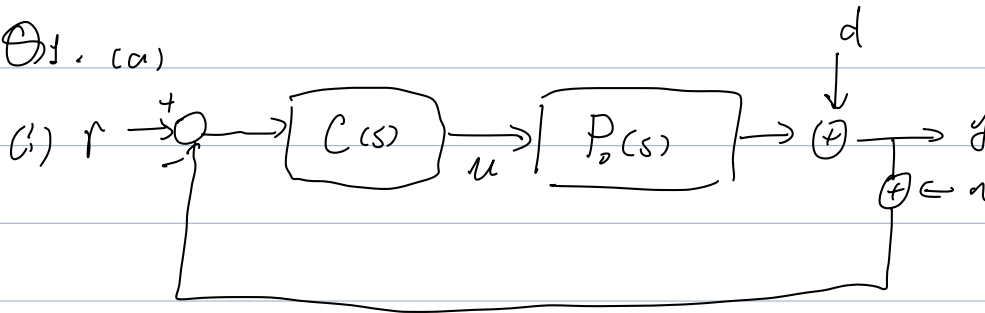


2024.

Q1. (a)



r : reference signal

u : control signal

d : disturbance signal

n : measurement noise

y : output signal.

A sensor is used to provide a measure of the plant output.

(ii)

* In terms of plant disturbance,

open loop control provides no compensation for disturbance

while close loop control can significantly attenuate

the disturbance error.

* In terms of possible changes in the plant gain

open loop control is highly sensitive to such changes

while the close loop control is less sensitive to them.

* In terms of stabilisation,

cost?

Open loop control may be unstable

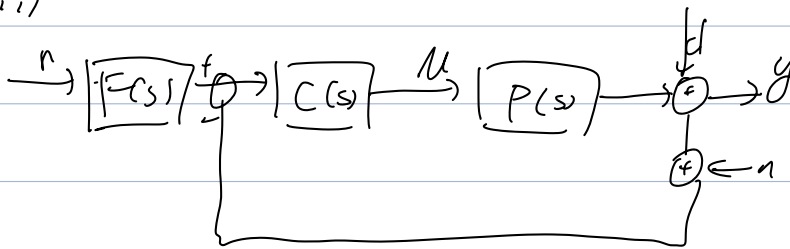
while close loop control is able to stabilise unstable system.

* However, close loop control may result in large control inputs to the plants. Therefore, the beneficial effects of feedback are limited by available plant capacity.

* Close loop control may also make the gain of the system too large that result in an unstable feedback system.

* Close loop control requires one sensor to measure the one, which can introduce measurement noise.

(ii)



The reference re-filter F allows the command response to be shaped independently of the feedback loop property.

(b).

$$G(s): \frac{Y(s)}{V(s)} = \frac{100}{s^2 + 23.8s + 810} \Rightarrow Y(s)s^2 + 23.8Y(s)s + 810Y(s) = 100V(s)$$

$$\Rightarrow \ddot{y} + 23.8\dot{y} + 810y = 100u$$

$$\text{Let } x_1 = y, \quad x_2 = \dot{y}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \text{then}$$

$$\dot{X} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -810 & -13.8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 100 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(c).

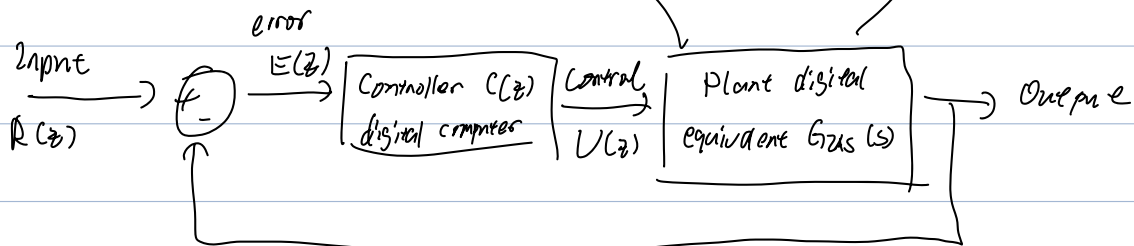
$$\begin{cases} \dot{x}_1 = -5.2x_1 - 8.3x_2 + 2u_1 \\ \dot{x}_2 = x_1 + 0.5u_2 \end{cases} \Rightarrow \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -5.2 & -8.3 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

$$\begin{cases} y_1 = 0.2x_2 \\ y_2 = 1.6x_1 \end{cases}$$

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 0.2 \\ 1.6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



Q2. (a).



(b) $\uparrow$ better approximation of continuous dynamics, faster smoother control action.

• larger stability margins and more accurate disturbance rejection

• But higher computational load, more data traffic, more sensitive to measurement noise.

Factors (a slow thermal process), processor speed

(c). $E = R - Y = (1 - G_{cl})R$.

$$\begin{aligned} e(\infty) &= \lim_{z \rightarrow 1} [(z-1)E(z)] \\ &= \lim_{z \rightarrow 1} [(z-1)(1 - G_{cl})R] \\ &= \lim_{z \rightarrow 1} [(z-1)(1 - \frac{L}{1+L})R] \end{aligned}$$

$$= \lim_{z \rightarrow 1} \left(\frac{z-1}{1+L} R \right)$$

$$\text{Let } L = \frac{N_L^*}{(z-1)^n D_L^*}$$

$$\text{then } e(\infty) = \lim_{z \rightarrow 1} \left(\frac{z-1}{1 + \frac{1}{(z-1)^n} \cdot \frac{N_L^*}{D_L^*}} R(z) \right)$$

$$= \lim_{z \rightarrow 1} \left(\frac{(z-1)^{n+1} D_L^*}{(z-1)^n D_L^* + N_L^*} R(z) \right)$$

$$\text{Let } R(z) = \frac{z}{z-1}, \text{ then}$$

$$e(\infty) = \lim_{z \rightarrow 1} \left(\frac{z(z-1)^{n+1} D_L^*}{(z-1)^n D_L^* + N_L^*} \right)$$

$$\text{therefore, } e(\infty) = \begin{cases} \frac{D_L^*}{D_L^* + N_L^*} = \frac{1}{1+L(1)}, & n=0 \\ 0, & n>0 \end{cases}$$

(d) The system type number n is the number of pure integrators in the open-loop transfer function.

$n=0$: finite non-zero steady-state error for step input.

$n \geq 1$: zero

Q3. (a), (1). 0.5 dB, 300 dB.

(ii) Cross-over frequency is the frequency where the magnitude of the transfer function equals 1, or 0dB.

10 rad/s

(iii) gain margin is infinite.

phase margin is 90° deg. at 10 rad/s

Delay margin is $90 \times \frac{\pi}{180} / 10 = 0.157 \text{ s}$.

(b). By increasing K_p , the magnitude of $L(j\omega)$ is scaled upward by a constant factor. This shifts the mag curve upwards on the Bode plot.

Therefore, the frequency where $|L(j\omega)| = 1$ increases.

$\Rightarrow \uparrow$ higher bandwidth.

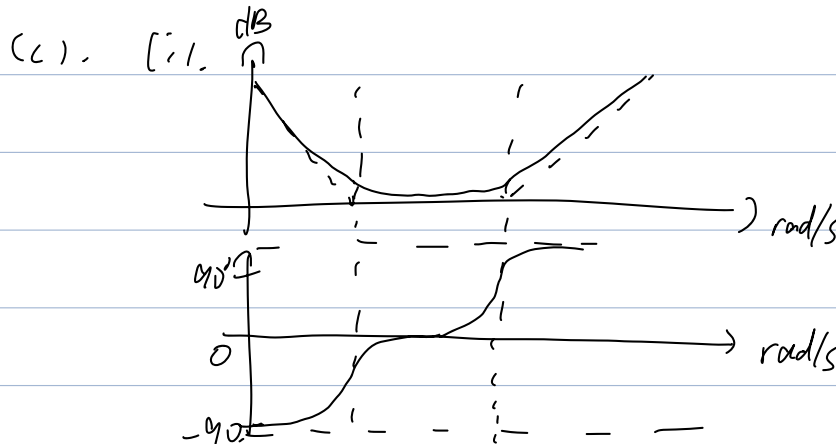
phase: it doesn't change the phase of $L(j\omega)$

Two limitations:

* Non-zero steady-state error

A P-only controller cannot eliminate e_{ss} without integrator

* Reduced stability, because it reduces phase margin



$$\begin{aligned}
 \text{(ii) From } C(s) &: K \left[1 + \frac{1}{T_I s} + T_D s \right] \\
 &= \frac{K}{s} \left(T_D s^2 + s + \frac{1}{T_I} \right) \\
 &\approx \frac{K}{s} \left[T_D s^2 + \left(1 + \frac{T_D}{T_I} \right) s + \frac{1}{T_I} \right] \\
 &\approx \frac{K}{s} (T_D s + 1) \left(s + \frac{1}{T_I} \right)
 \end{aligned}$$

Since the number of zeros is larger than the number of poles, this sys is not realisable.

(Current output depends on future input values).

To make it realisable:

$$C_r = K \frac{(T_D s + 1) \left(s + \frac{1}{T_D} \right)}{s (T_D s + 1)}$$

$$\text{(iii). } \frac{1}{3.00}, \frac{1}{0.4}, \quad K = 10 \text{ to } 25. \dots$$

$$\begin{aligned}
 \text{Q5. (a). } C(z) &= G_{ZAS}(z) \\
 &= \frac{L(z)}{1 + L(z)}
 \end{aligned}$$

(b). type number $n=1$.

$$G_{ce} = \frac{K L(z)}{1 + L(z)}$$

$$C(z) = (1 - G_{ce}) R(z).$$

$$= \frac{1}{1+KL(z)} R(z).$$

$$= \frac{z-1}{(z-1) + \frac{1.5K}{z+0.2}} \cdot \frac{z}{z-1} \quad (\text{stop input } 1,$$

$$= \frac{z}{(z-1) + \frac{1.5K}{z+0.2}}$$

$$e_{ss} = \lim_{z \rightarrow 1} (z-1) \cdot \frac{z}{(z-1) + \frac{1.5K}{z+0.2}} = 0$$

$$(c), \quad \zeta = 0.4 = \frac{1}{3 \omega_n}$$

$$G_{cl} = \frac{KL}{1+KL} = \frac{1.5K}{z^2 - 0.8z + 1.5K - 0.2}$$

$$z^2 - 0.8z + 1.5K - 0.2 = z^2 - 2\cos(\omega_d T) e^{-\zeta \omega_n T} z + e^{-2\zeta \omega_n T}$$

$$\Rightarrow \begin{cases} 2\cos(\omega_d T) e^{-\zeta \omega_n T} = 0.8 \\ e^{-2\zeta \omega_n T} = 1.5K - 0.2 \end{cases}$$

$$\frac{1}{T} \cos^{-1}\left(\frac{0.8}{2} e^{\frac{T}{\tau}}\right)$$

$$\zeta = \frac{1}{3 \omega_n} \Rightarrow \begin{cases} 2\cos \omega_d T \cdot e^{-\frac{T}{\tau}} = 0.8 \\ e^{-\frac{2T}{\tau}} = 1.5K - 0.2 \end{cases}$$

$$\Rightarrow \begin{cases} K = 0.5377 \\ K = \cancel{0.3859} \end{cases}$$

$$K = \frac{e^{-\frac{T}{\tau} + 0.2}}{1.5}$$

$$\omega_d = \cancel{73.146} \text{ rad/s} = \omega_n \sqrt{1-\zeta^2} = \frac{\sqrt{1-\zeta^2}}{\zeta \tau}$$

$$\Rightarrow \zeta^2 = \frac{1-\zeta^2}{\omega_d^2} \Rightarrow \zeta = \sqrt{\frac{1}{\omega_d^2 \tau^2 + 1}}$$

$$\Rightarrow \zeta = \sqrt{\frac{1}{(\tau \omega_d)^2 + 1}} = 0.236$$

$$\begin{aligned}
 (d). e(z) &= \left(1 - \frac{1.5K}{z^2 - 0.8z + 1.5K - 0.2} \right) \cdot \frac{Tz}{(z-1)^2} \\
 &= \frac{z^2 - 0.8z - 0.2}{z^2 - 0.8z + 1.5K - 0.2} \cdot \frac{Tz}{(z-1)^2}
 \end{aligned}$$

$$e_{ss} = \lim_{z \rightarrow 1} (z-1) \cdot e(z)$$

$$= \lim_{z \rightarrow 1} \frac{Tz(z-0.2)}{z^2 - 0.8z + 1.5K - 0.2}$$

$$= \frac{0.1 \times (1+0.2)}{1 - 0.8 + 1.5K - 0.2}$$

$$= \frac{0.12}{1.5K} = 0.1428$$

(e).

Q6.

$$(a). P_{ZAS}(z) = (1 - z^{-1}) \cdot Z\left(\frac{P(s)}{s}\right)$$

$$= (1 - z^{-1}) \cdot Z\left(\frac{1}{s^2(s-0.1)}\right)$$

$$= (1 - z^{-1}) \cdot Z\left(\frac{A}{s^2} + \frac{B}{s} + \frac{C}{s-0.1}\right)$$

$$A(s-0.1) + Bs(s-0.1) + Cs^2$$

$$\Rightarrow (B+C)s^2 + (A-0.1B)s - 0.1A$$

$$\Rightarrow \begin{cases} A = -10 \\ B = -100 \\ C = 100 \end{cases}$$

$$\begin{aligned} P_{ZAS}(z) &= (1-z^{-1}) \cdot z \left(\frac{-10}{s^2} + \frac{-100}{s} + \frac{100}{s-0.1} \right) \\ &= \left(\frac{z-1}{z} \right) \cdot \left(-10 \cdot \frac{z}{(z-1)^2} - 100 \cdot \frac{z}{z-1} + 100 \cdot \frac{z}{z-e^{0.1T}} \right) \\ &= -10 \cdot \frac{1}{z-1} - 100 + 100 \frac{z-1}{z-1.10517} \end{aligned}$$

$$\begin{aligned} &\underbrace{100 \frac{z-1 - z + 1.10517}{z-1.10517}} \\ &\quad \frac{10.517}{z-1.10517} \end{aligned}$$

$$\begin{aligned} &= \frac{-10(z-1.10517) + 10.517(z-1)}{(z-1)(z-1.10517)} \\ &= \frac{0.517z + 0.533}{(z-1)(z-1.10517)} \\ &= 0.517 \frac{z + 1.031}{(z-1)(z-1.10517)} \end{aligned}$$

$$(b). \quad \underbrace{n=1}_{\text{}} \quad e_{ss} = \lim_{z \rightarrow 1} (1-z^{-1}) \cdot \left(1 - \frac{P(z)}{1+P(z)} \right) \cdot R(z).$$

$$= \lim_{z \rightarrow 1} \frac{z - 1.105}{0.517 (z + 1.031)} \cdot T$$

$$= -0.1 \neq 0$$

NO.

(c).

$$p1. \quad K \frac{z-1}{z-1}$$

$$\frac{1}{(1+F(z))} \cdot \frac{\cancel{z-1} \cdot T \cdot \cancel{z-1}}{\cancel{z-1} \cdot (z-1)}$$

$$N(z) = 0 \quad \text{ess} = 0$$

(d).

$$(cs) \Rightarrow \frac{Y(s)}{U(s)} = \frac{s-0.5}{s} \Rightarrow sY(s) = sU(s) - 0.5U(s)$$

$$\Rightarrow \dot{y}(t) = \dot{u}(t) - 0.5u(t)$$

$$\Rightarrow \frac{y_{k+1} - y_k}{T} = \frac{u_{k+1} - u_k}{T} - 0.5u_k$$

$$\Rightarrow Y(z) \cdot z - Y(z) = U(z)z - U(z) - 0.5U(z)T$$

$$\Rightarrow \frac{Y(z)}{U(z)} = \frac{z-1-0.5T}{z-1}$$